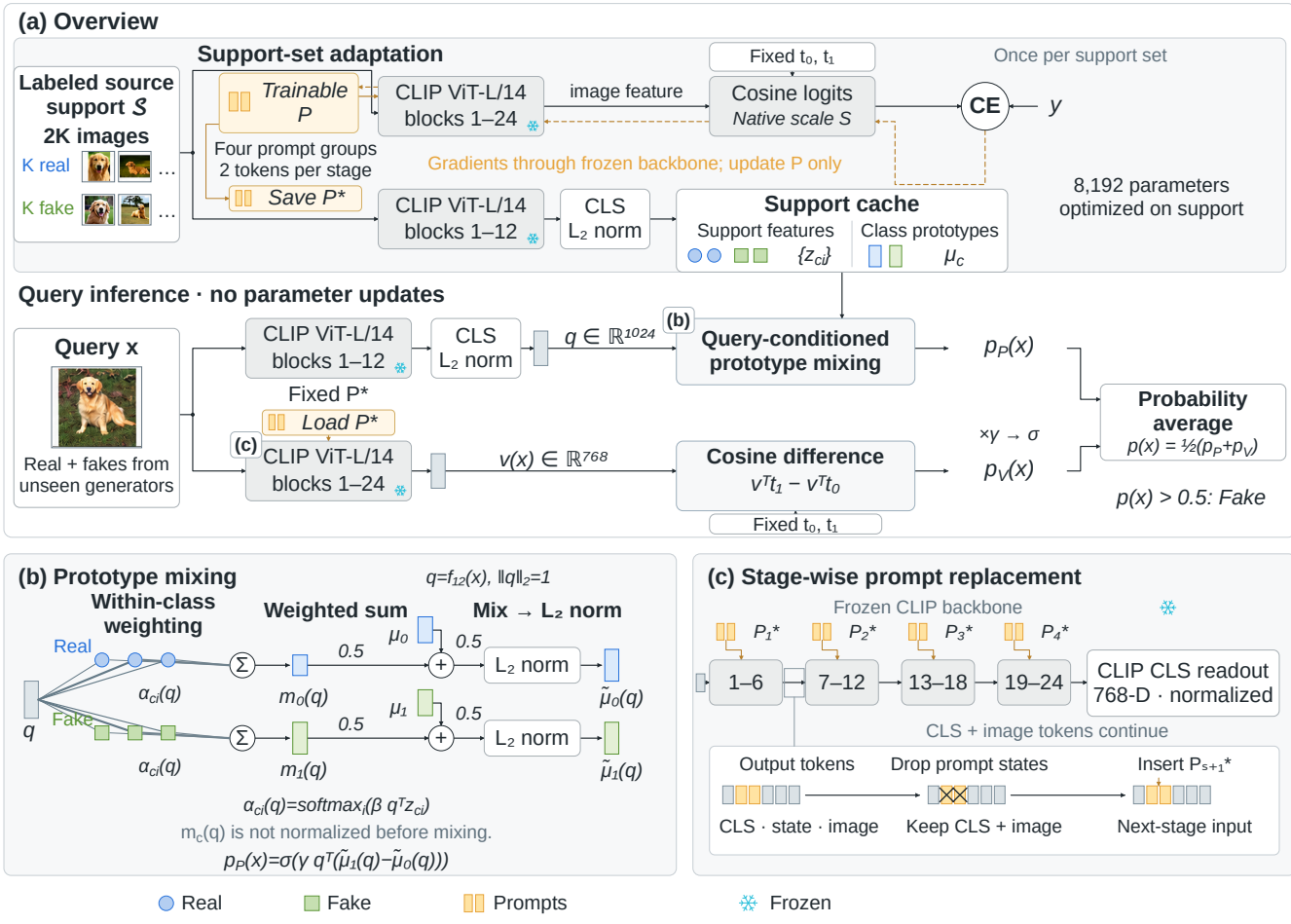

(a) Overview
Support-set adaptation
Labeled source support S
2K images
K real
K fake
Trainable P
Four prompt groups
2 tokens per stage
Save P*
CLIP ViT-L/14 blocks 1–24
CLIP ViT-L/14 blocks 1–12
image feature
Fixed t₀, t₁
Cosine logits
Native scale S
CE
y
Once per support set
Gradients through frozen backbone; update P only
CLS
L₂ norm
Support cache
Support features
{z_ci}
Class prototypes
μ_c
8,192 parameters
optimized on support
Query inference · no parameter updates
Query x
Real + fakes from unseen generators
CLIP ViT-L/14 blocks 1–12
Fixed P*
Load P*
(c)
CLIP ViT-L/14 blocks 1–24
CLS
L₂ norm
q ∈ ℝ^1024
(b)
Query-conditioned prototype mixing
p_P(x)
v(x) ∈ ℝ^768
Cosine difference
v^T t₁ − v^T t₀
Fixed t₀, t₁
×γ → σ
p_V(x)
Probability average
p(x) = ½(p_P+p_V)
p(x) > 0.5: Fake
(b) Prototype mixing
q=f₁₂(x), ‖q‖₂=1
Within-class weighting
Weighted sum
Mix → L₂ norm
Real
Fake
q
α_ci(q)
m₀(q)
m₁(q)
μ₀
μ₁
0.5
L₂ norm
m̃₀(q)
m̃₁(q)
α_ci(q)=softmax_i(β q^T z_ci)
m_c(q) is not normalized before mixing.
p_P(x)=σ(γ q^T(μ̃₁(q)−μ̃₀(q)))
(c) Stage-wise prompt replacement
Frozen CLIP backbone
P₁*
P₂*
P₃*
P₄*
1–6
7–12
13–18
19–24
CLIP CLS readout
768-D · normalized
CLS + image tokens continue
Output tokens
Drop prompt states
Insert P_s+1*
CLS · state · image
Keep CLS + image
Next-stage input
Real
Fake
Prompts
Frozen